

Adaptations for acquiring resources were key steps in the evolution of vascular plants

EVOLUTION Most plants grow in soil and therefore inhabit two worlds—above ground, where shoots acquire sunlight and CO_2 , and below ground, where roots acquire water and minerals. The successful colonization of the land by plants depended on adaptations that allowed early plants to acquire resources from these two different settings.

The algal ancestors of plants absorbed water, minerals, and CO_2 directly from the water in which they lived. Transport in these algae was relatively simple because every cell was close to the source of these substances. The earliest plants were nonvascular and produced photosynthetic shoots above the shallow fresh water in which they lived. These leafless shoots typically had waxy cuticles and few stomata, which allowed them to avoid excessive water loss while still permitting some exchange of CO_2 and O_2 for photosynthesis. The anchoring and absorbing functions of early plants were assumed by the base of the stem or by threadlike rhizoids (see Figure 29.16).

As plants evolved and increased in number, competition for light, water, and nutrients intensified. Taller plants with broad, flat appendages had an advantage in absorbing light. This increase in surface area, however, resulted in more evaporation and therefore a greater need for water. Larger shoots also required stronger anchorage. These needs favored the production of multicellular, branching roots. Meanwhile, as greater shoot heights further separated the top of the photosynthetic shoot from the nonphotosynthetic parts below ground, natural selection favored plants capable of efficient long-distance transport of water, minerals, and products of photosynthesis.

The evolution of vascular tissue consisting of xylem and phloem made possible the development of extensive root and shoot systems that carry out long-distance transport (see Figure 35.10). The **xylem** transports water and minerals from roots to shoots. The **phloem** transports products of photosynthesis from where they are made or stored to where they are needed. **Figure 36.2** provides an overview of resource acquisition and transport in an actively photosynthesizing plant.

Shoot Architecture and Light Capture

Because most plants are photoautotrophs, their success depends ultimately on their ability to photosynthesize. Over the course of evolution, plants have developed a wide variety

▼ **Figure 36.2** An overview of resource acquisition and transport in a vascular plant during the day.

In leaves and green stems, gas exchange occurs through stomata. During photosynthesis, CO_2 is taken up and O_2 is produced. Cells use some of that O_2 for cellular respiration, and the rest is released into the air.

Transpiration, the loss of water from leaves (mostly through stomata), creates a force within leaves that pulls xylem sap upward.

Water and minerals are transported upward from roots to shoots as xylem sap.

Water and minerals in the soil are absorbed by roots.

H_2O

CO_2

O_2

Sugar

Light

Sugars are produced by photosynthesis in the leaves.

Phloem sap can flow both ways between shoots and roots. It moves from sites of sugar production (usually leaves) or storage (usually roots) to sites of sugar use or storage.

O_2
 CO_2

As root cells carry out their cellular respiration, they exchange gases with the air spaces of the soil, taking in O_2 and discharging CO_2 .

MAKE CONNECTIONS How would you change the figure in order to show gas exchanges at night? (See Figure 10.22.)